

Yingtian “Bill” Chen

陈颖天 · 陳穎天

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Employment

- **OSU Center for Cosmology and AstroParticle Physics** | Columbus, OH, US Since 2026
CCAPP Postdoctoral Fellow

Education

- **University of Michigan Department of Astronomy** | Ann Arbor, MI, US 2020 – 2026
Ph.D. and M.S. in Astronomy and Astrophysics
- **Peking University School of Physics** | Beijing, China 2016 – 2020
B.S. in Physics (with honours)

Experience

- **MIT Kavli Institute** | Cambridge, MA, US 2019
Visiting Researcher

Research Interests

Star clusters · Stellar streams · Galaxy formation · Galactic Archaeology · Computational astronomy

- Probing galaxy structure and star cluster evolution via stellar streams.
- Simulating galaxy and star cluster formation in cosmological contexts.
- Semi-analytical modeling of star cluster evolution in galactic environment.

Honours and Awards

- **CCAPP Postdoctoral Fellowship**, OSU 2026
- **Rackham Conference Travel Grant** × 3, UM 2023 – 2026
- **Rackham Predoctoral Fellowship**, UM 2025
- **Rackham International Student Fellowship**, UM 2021
- **Weiming Physics Scholarship**, PKU 2020
- **Outstanding Graduate** of Beijing 2020
- **First Prize + Best speaker**, Xingcheng Forum, PKU 2019
- **Huabao Funding for Undergraduate Research Program**, PKU 2018
- **National Scholarship** 2018
- **Pacemaker to Merit Student**, PKU 2018
- **Outstanding Winner + SIAM Award**, Mathematical Contest in Modeling 2018
- **Gold Medal**, Chinese Physics Olympiad 2015

Awarded Resources

Observational Resources

- **PI**, *Roman* Cycle 1, ID: 19072, 150,000 USD Since 2027
- **Co-PI**, Magellan/M2FS 2026A, 2 nights 2026

Computational Resources

- **Co-PI**, ACCESS *Maximize* AST200017, awarded allocation $\approx$ 50,000 USD 2025 – 2027
171K SUs ($\approx$ 10M CPU hours) @ TACC/Stampede3 + 256 TB @ TACC/Ranch
- **Co-PI**, ACCESS *Accelerate* AST200017, awarded allocation $\approx$ 30,000 USD 2024 – 2025
116K SUs ($\approx$ 6M CPU hours) @ TACC/Stampede3 + 100 TB @ TACC/Ranch

Press Releases

- Michigan News, *Talk about streaming bundles: U-M astronomers discover 87 stellar stream candidates in the Milky Way* [[link](#)] 2026
- AAS Journal Author Series, *Yingtian (Bill) Chen on 2025ApJ...995...15C* [[link](#)] 2025
- Phys.org, *Astronomers identify dozens of stellar streams with Gaia* [[link](#)] 2025
- THU, *First direct measurements of globular cluster mass loss through stellar streams* [[link](#)] 2025

Selected Talks

- **Conference Talk**, *CCAPP Fellows Symposium*, OSU | Columbus, OH, US 2026
- **Invited Seminar**, *CCAPP seminar*, OSU | Columbus, OH, US 2026
- **Dissertation Talk**, *AAS 247* | Phoenix, AZ, US 2026
- **Invited Talk**, *S5 telecon* | Remote 2025
- **Invited Talk** × 2, *DESI MWS telecon* | Remote 2024 – 2025
- **Invited Seminar**, *Carnegie Observatories Stream Team meeting* | Remote 2025
- **Invited Seminar**, *lunch talk*, Penn State | University Park, PA, US 2025
- **Conference Talk + Poster**, *Gravity in the Local Group*, CMU | Pittsburgh, PA, US 2025
- **Invited Seminar**, *KICP seminar*, UChicago | Chicago, IL, US 2025
- **Invited Seminar**, *Nearby Universe group meeting*, Flatiron Institute | New York, NY, US 2025
- **Invited Seminar**, *American Museum of Natural History* | New York, NY, US 2025
- **Lunch Talk** × 5, *Astronomy grad lunch talk series*, UM | Ann Arbor, MI, US 2021 – 2025
- **Poster + Flash Talk**, *DGSCS 2024*, UChicago | Chicago, IL, US 2024
- **Invited Seminar**, PKU · THU · SHNU · SHAO · SJTU · PMO · NJU · ZJU | Beijing · Shanghai · Nanjing · Hangzhou, China 2024
- **Invited Seminar**, *Galaxy Formation seminar*, Flatiron Institute | New York, NY, US 2024
- **Invited Talk**, UChicago | Remote 2024
- **Invited Talk**, *SMWLVS Star Clusters Working Group meeting* | Remote 2023
- **Conference Talk**, *MODEST-23*, NU | Evanston, IL, US 2023
- **Conference Talk**, *Great Lakes Clusters and Streams*, UM | Ann Arbor, MI, US 2023
- **Talk**, *Seminar for undergraduate students*, PKU | Beijing, China 2019
- **Talk**, *Xingcheng Forum*, PKU | Beijing, China 2019
- **Talk**, *Seminar for theoretical physics*, FDU | Shanghai, China 2019

Student Supervision

- Colin Holm-Hansen, *Generating mock catalogs of stellar streams in simulated Milky Way-like galaxies* (co-supervised with Oleg Gnedin), UM grad student 2024 – 2026
- Brigitte Vazquez Segovia, *Modeling subhalo interactions in the GD-1 stellar stream* (co-supervised with Monica Valluri), UM grad student 2024 – 2026

Professional Service

- **Referee of Journal Articles** (8), A&A (3) · ApJ (1) · MNRAS (3) · OJAp (1) Since 2023
- **Conference Session Chair**, *DGSCS 2024*, UChicago | Chicago, IL, US 2024
- **Conference LOC Chair**, *Great Lakes Clusters and Streams*, UM | Ann Arbor, MI, US 2023
- **Organizer**, *Stellar Halos Group meeting* (weekly), UM | Ann Arbor, MI, US 2024 – 2025

Code Development

- **Author**, [StarStream](#), Automatic detection method for stellar streams Since 2025
- **Author**, [GC_formation_model](#), Semi-analytical model of globular cluster formation Since 2023
- **Author**, [prj_plotter](#), Density projection tool for simulations using quadtree Since 2022
- **Author**, [mesh_illustris](#), Fast toolkit to analyze Illustris data with mesh Since 2021
- **Contributor**, [gala](#), Python package for galactic and gravitational dynamics Since 2024
- **Contributor**, [galax](#), JAX-based package for galactic and gravitational dynamics Since 2024
- **Contributor**, [galpy](#), Python package for galactic dynamics Since 2024
- **Contributor**, [ART](#), Simulation code using adaptive mesh refinement Since 2023
- **Public Software**, [fov_simulator](#), Simulator of camera field of view for astrophotography Since 2025

University Service

- **Organizer**, Astronomy grad lunch talk series (weekly) 2024 – 2026
- **Chair**, UM Chinese astronomers networking group 2022 – 2026
- **Organizer**, Build-Your-Website workshop 2025
- **Organizer**, Astrocoffee journal club (bi-weekly) 2022 – 2025
- **Organizer**, Preliminary examination preparation club (weekly) 2022 – 2025
- **Department Bread Baker** (weekly) 2021 – 2022

Outreach

- **Talk**, How scientists find black holes, UM Museum of Natural History | Ann Arbor, MI, US 2025
- **Seminar**, Reductionism and holism, Yuexia Space seminar series | Shenzhen, China 2021
- **Guest Lecturer**, Physics Olympiad training, Chengdu Jinjiang Jiaxiang Foreign Language Senior High School | Chengdu, China 2017
- **Online Tutorial**, 2D Ising model in Matlab and Python, published on GitHub and Zhihu 2019
- **Online Tutorial**, BP neural network in Matlab, published on GitHub 2018

Teaching

- **Guest Lecturer**, Hamiltonian Mechanics, UM | Ann Arbor, MI, US 2025
Galactic Dynamics Summer Workshop
- **Guest Lecturer**, Cosmological N-body Simulations, UM | Ann Arbor, MI, US 2025
ASTRO 534: The Extragalactic Universe (grad-level cosmology)
- **Teaching Assistant**, UM | Ann Arbor, MI, US 2021 & 2024
ASTRO 104: Alien Skies: A Tour Through the Universe
ASTRO 106: Aliens
ASTRO 115: Introductory Astrobiology: The Search for Life in the Universe

Skills

High performance computing · Data analysis and visualization · Machine learning · Web development

- **Programming Languages:** C/C++ · Python · Latex · MATLAB · HTML/CSS
- **Software/Packages:** ART · AREPO · GADGET · PeTar · MPI · AGAMA · multiprocessing · NumPy · Matplotlib · SciPy · scikit-learn · PyTorch · Astropy · yt · gala · galax · galpy · Bootstrap · Git
- **Languages:** Mandarin Chinese (native) · English (fluent)

Publications

- 16 in total = 11 as first author + 1 by (co-)supervised students + 4 as contributing author
- First-author citations > 220 · h-index = 8
- Total citations > 260 · h-index = 9

Publications as First Author

1. **Yingtian Chen**[✉], Oleg Y. Gnedin, Hui Li, Vadim A. Semenov, and Mark Gieles, *SCARF: Simulating clustered star formation in the early Universe*, in prep. 2026
2. **Yingtian Chen**[✉], Oleg Y. Gnedin, and Adrian M. Price-Whelan, *StarStream on Gaia: Stream discovery and mass loss rate of globular clusters*, *ApJS* **283**, 60 [ApJS 283, 60](#) 2026
3. **Yingtian Chen**[✉], Oleg Y. Gnedin, Adrian M. Price-Whelan, and Colin Holm-Hansen, *StarStream: Automatic detection algorithm for stellar streams*, *ApJ* **995**, 15 2025
4. **Yingtian Chen**[✉], Hui Li[✉], and Oleg Y. Gnedin, *Stellar streams reveal the mass loss of globular clusters*, *ApJL* **980**, L18 2025
5. **Yingtian Chen**[✉], Monica Valluri, Oleg Y. Gnedin, and Neil Ash, *Improved particle spray algorithm for modeling globular cluster streams*, *ApJS* **276**, 32 2025
6. **Yingtian Chen**[✉] and Oleg Y. Gnedin, *Galaxy assembly revealed by globular clusters*, *OJAp* **7** 2024
7. **Yingtian Chen**[✉] and Oleg Y. Gnedin, *Catalogue of model star clusters in the Milky Way and M31 galaxies*, *MNRAS* **527**, 3692 2024
8. **Yingtian Chen**[✉] and Oleg Y. Gnedin, *Formation of globular clusters in dwarf galaxies of the Local Group*, *MNRAS* **522**, 5638 2023
9. **Yingtian Chen**[✉] and Oleg Y. Gnedin, *Modeling the kinematics of globular cluster systems*, *MNRAS* **514**, 4736 2022
10. **Yingtian Chen**, Hui Li[✉], and Mark Vogelsberger, *Effects of initial density profiles on massive star cluster formation in giant molecular clouds*, *MNRAS* **502**, 6157 2021
11. **Yingtian Chen** and Bo-Qiang Ma[✉], *Novel pre-burst stage of gamma-ray bursts from machine learning*, *JHEAp* **32**, 78 2021

Publications by *(Co-)supervised Students

12. *Colin Holm-Hansen[✉], **Yingtian Chen**, and Oleg Y. Gnedin, *Catalog of Mock Stellar Streams in Milky Way-like Galaxies*, *OJAp* **9** 2025

Publications as Contributing Author

13. Brandon Sike[✉], Mateusz Ruszkowski, Oleg Y. Gnedin, **Yingtian Chen**, et al., *Resolving star cluster formation in galaxy simulations with cosmic ray feedback*, *ApJ* **1004**, 236 2025
14. Micheli T. Moura[✉], Ana L. Chies-Santos, Cristina Furlanetto, **Yingtian Chen**, et al., *Tracing the relic nature of compact galaxies through their globular cluster systems*, *ApJ* **976**, 189 2026
15. Neil Ash[✉], Monica Valluri, **Yingtian Chen**, and Eric F. Bell, *Stellar bars form dark matter counterparts in TNG50*, *ApJ* **976**, 189 2024
16. Sarah Pearson[✉], Ana Bonaca, **Yingtian Chen**, and Oleg Y. Gnedin, *Forecasting the population of globular cluster streams in Milky Way-type galaxies*, *ApJ* **976**, 54 2024